RETAIL ANALYTICS IN ACTION: MODEL COMPARISON FOR BUSINESS OPTIMIZATION

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Executive Summary:

This study aimed to improve Fukushimart's operational efficiency by addressing three critical problems: manual inventory tracking, ineffective customer inquiry handling, and the absence of predictive tools for demand forecasting. To tackle these challenges, seven machine learning models were implemented and evaluated: Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Support Vector Machine (SVM), Naive Bayes, Linear Regression, and Polynomial Regression. Among these models, KNN achieved perfect accuracy (100%) in classifying best selling items but showed limitations in scalability and risk of overfitting. SVM followed closely with a 97.3% accuracy rate, offering a more practical and scalable alternative for real-time applications. For predicting customer behavior, Logistic Regression demonstrated the highest accuracy at 95%, making it the most reliable model for identifying purchase intent. In forecasting monthly inventory restocking, Polynomial Regression significantly outperformed Linear Regression, achieving an R^2 score of 98.5% and a low RMSE of 0.25, indicating its superior ability to handle non-linear trends in sales and stock patterns. Overall, these results show that machine learning can effectively support automated decision-making, optimize inventory management, and improve customer engagement at Fukushimart.

Introduction:

In the evolving landscape of retail and e-commerce, data-driven strategies are imperative to remain competitive. Fukushimart, a convenience store integrating virtual shopping and real-time analytics, aims to transition from manual practices to intelligent predictive systems. This project integrates machine learning to address challenges in customer purchase forecasting and inventory control using a structured analytical framework.

Statement of the Problem:

Fukushimart faces three critical issues:

1. Manual inventory tracking that delays restocking and causes stockouts.
2. Ineffective handling of customer inquiries, resulting in lost sales opportunities.
3. Lack of predictive capabilities for identifying high-demand items or bulk purchase tendencies.

These issues lead to inefficiencies in store operations, reduce customer satisfaction, and hinder profitability.

Objectives of the Analysis:

- To predict customer purchase behavior.
- To optimize inventory management through predictive modeling.
- To recommend efficient operational practices based on analytical insights.
- To compare model performance and accuracy for data-driven decision-making.

Data Description:

The dataset consists of a 3-year transaction and inventory records from Fukushimart's virtual store. Key attributes include:

- **Customer Behavior:** Action (Buy/Ask), Response Time
- **Sales:** Final Spend, Quantity, Item
- **Inventory:** Price, Stock Levels, and Units Sold per Month

Data was preprocessed to handle missing values, encode categorical variables, and normalize numeric features.

Methodology:

This analysis uses a supervised machine learning approach. The dataset was split into training and testing sets. Each model was trained, validated, and tested using standard procedures:

1. Preprocessing via normalization and encoding
2. Model training using Scikit-learn
3. Evaluation using metrics such as Accuracy, Precision, Recall, F1-score, and Support for the Classification Models. While R^2 Score and Root Mean Square Error (RMSE) was used for the Regression Models.
4. Visualization using matplotlib and seaborn for interpretability

Exploration Data Analysis: EDA performs all data normalization during the creation of all the models.

EDA was conducted to uncover trends and prepare data for modeling:

- Outliers were identified and retained to preserve realistic transaction variance
- FinalSpend and Quantity were binned for classification analysis
- All features were normalized during model construction

In-Depth Analysis:

Each of the seven models was implemented and evaluated for two primary tasks:

- Classification: Predicting purchase intent and identifying bestsellers
- Regression: Estimating stock quantity needed monthly

Data Representation: Graphical Presentation of Results

1. Best Seller Prediction

1.1 Decision Tree

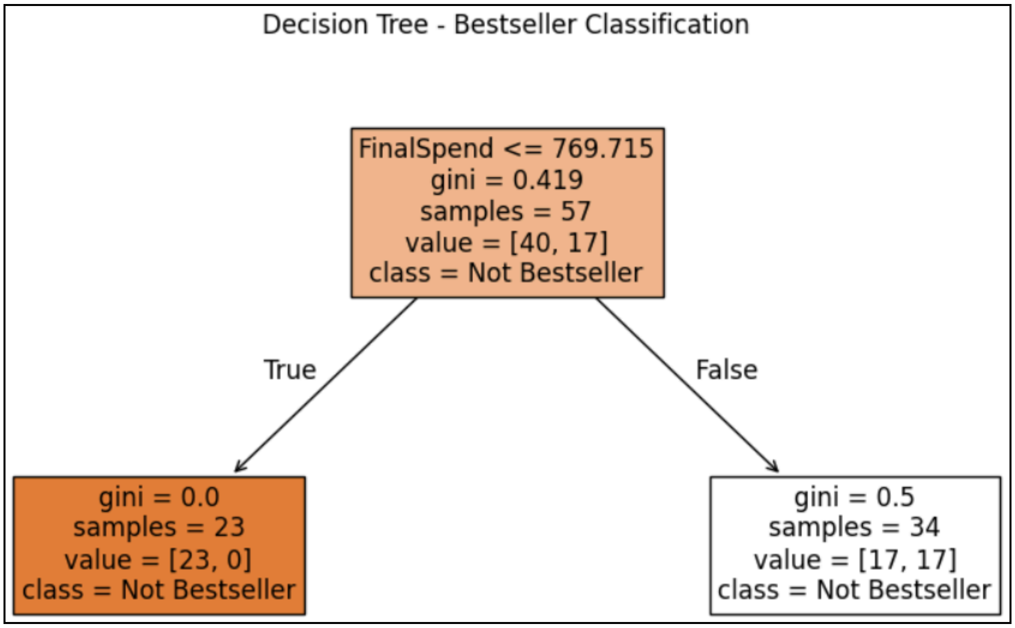

Figure 1.1.1 Decision Tree 1

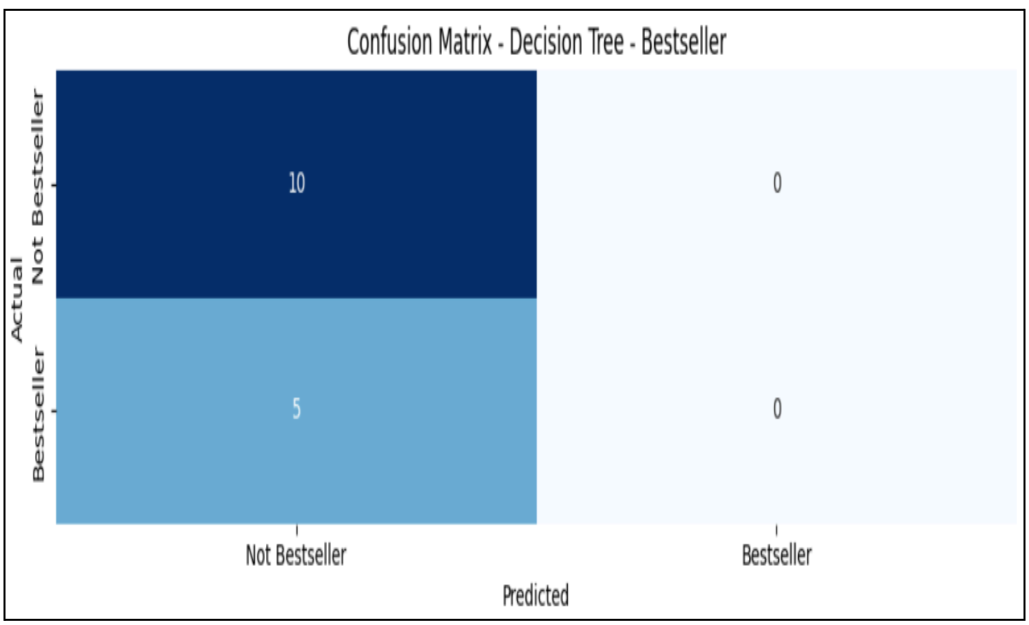

Figure 1.1.2 Decision Tree Regression Confusion Matrix 1

	Precision	**Recall**	**F1- Score**	**Support**
0	0.67	1.00	0.80	10
1	0.00	0.00	0.00	5

Accuracy			0.67	15
Macro Avg	0.33	0.50	0.40	15
Weighted Avg	0.44	0.67	0.53	15

Table 1.1.1 Decision Tree Classification Report 1

Fold 1:	0.66
Fold 2:	0.73
Fold 3:	0.71
Fold 4:	0.71
Fold 5:	0.71
Average Accuracy:	0.66 or **66%**

Table 1.1.2 Decision Tree Fold Cross-Validation Accuracy Scores 1

1.2 Logistic Regression

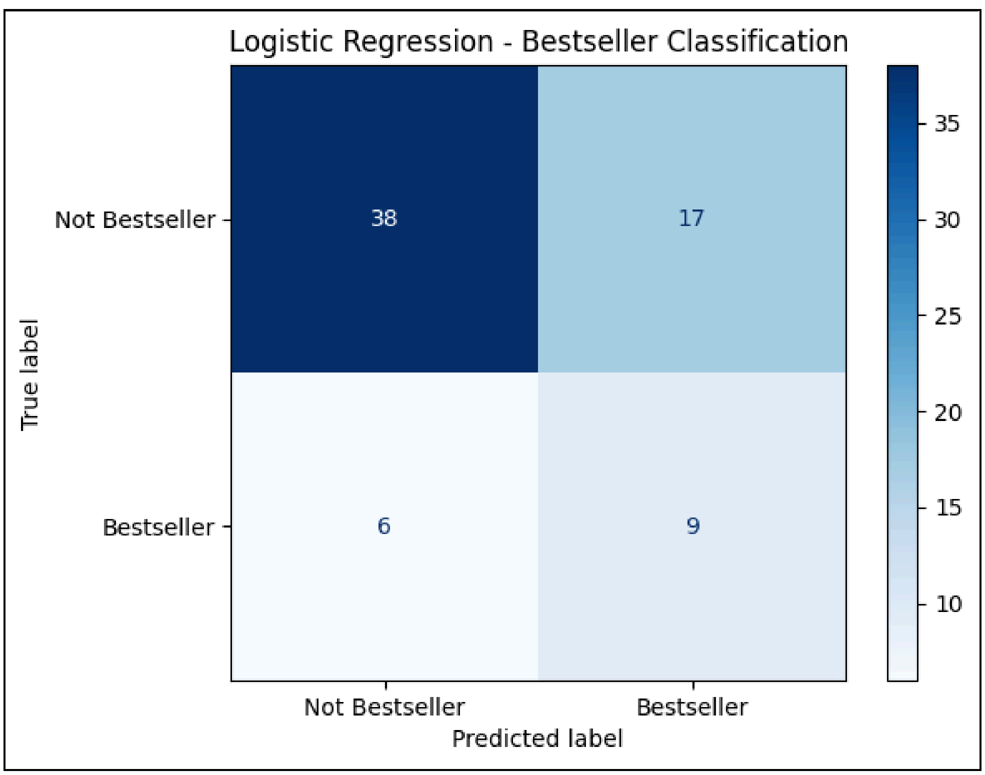

Figure 1.2.1 Logistic Regression Confusion Matrix 1

	Precision	**Recall**	**F1- Score**	**Support**
0	0.86	0.69	0.77	55
1	0.35	0.60	0.44	15
Accuracy			0.67	70
Macro Avg	0.60	0.65	0.60	70
Weighted Avg	0.75	0.67	0.70	70

Table 1.2.1 Logistic Regression Classification Report 1

Fold 1:	0.60
Fold 2:	0.59
Fold 3:	0.59
Fold 4:	0.75
Fold 5:	0.64
Average Accuracy:	0.63 or **63%**

Table 1.2.2 Logistic Regression 5 Fold Cross-Validation Accuracy Scores 1

1.3 KNN

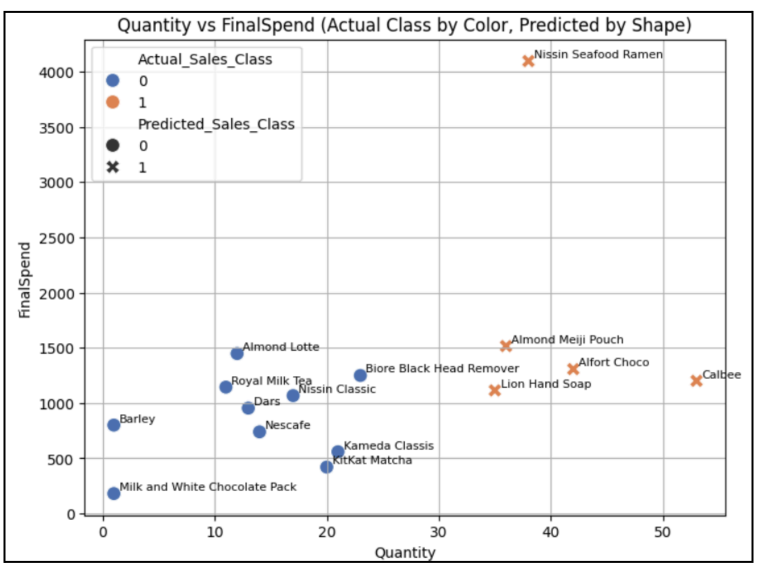

Figure 1.3.1 KNN 1

	Precision	**Recall**	**F1- Score**	**Support**
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	5
Accuracy			1.00	15
Macro Avg	1.00	1.00	1.00	15
Weighted Avg	1.00	1.00	1.00	15

Bootstrap Mean Accuracy:	0.9867 or 98.67%
Standard Deviation :	+_ 0.0321 or 3.21%

Table 1.3.1 KNN Classification Report 1

Fold 1:	1.00
Fold 2:	0.93
Fold 3:	1.00
Fold 4:	1.00
Fold 5:	1.00
Average CV Accuracy:	98.67%

Table 1.3.2 KNN 5 Fold Cross-Validation Accuracy Scores 1

1.4 Naive Bayes

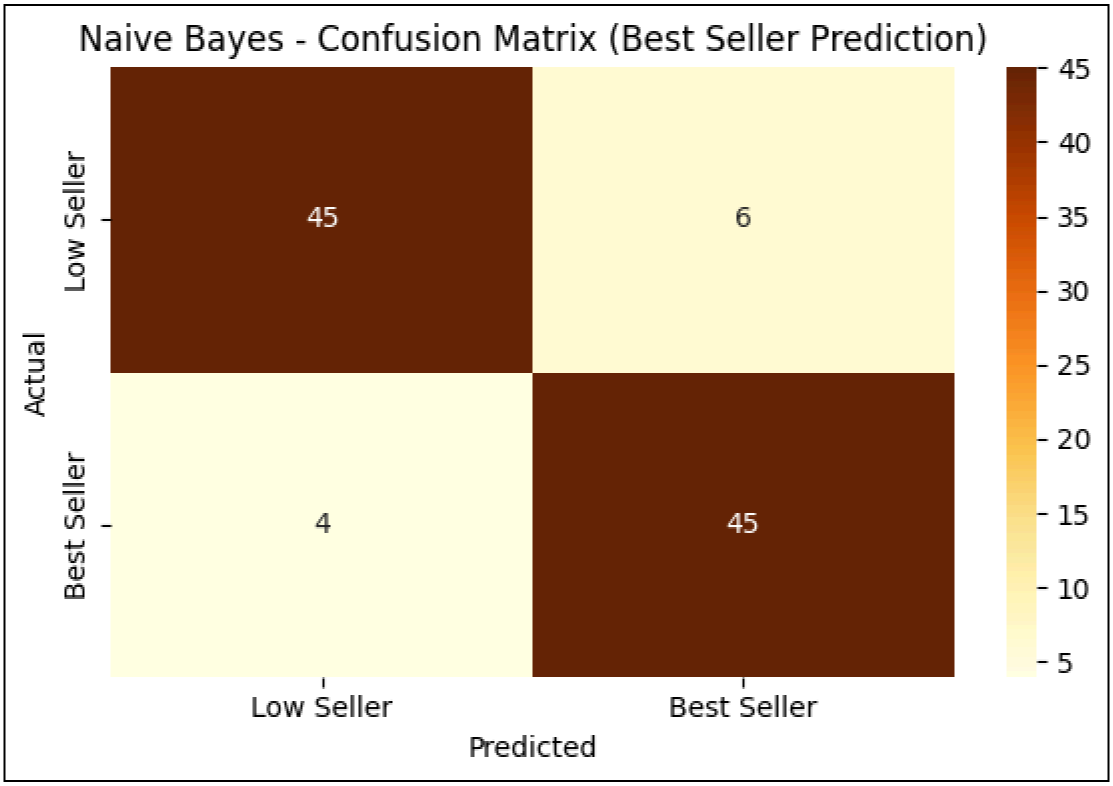

Figure 1.4.1 Naive Bayes Confusion Matrix 1

	Precision	**Recall**	**F1- Score**	**Support**
0	0.92	0.88	0.90	51
1	0.88	0.92	0.90	49
Accuracy			**0.90**	100
Macro Avg	0.90	0.90	0.90	100
Weighted Avg	0.90	0.90	0.90	100

Table 1.4.1 Naive Bayes Classification Report 1

Fold 1:	0.94
Fold 2:	0.94
Fold 3:	0.88

Fold 4:	0.92
Fold 5:	0.87
Average Accuracy:	0.91 or **91%**

Table 1.4.2 Naive Bayes 5 Fold Cross-Validation Accuracy Scores 1

1.5 Support Vector Machine (SVM)

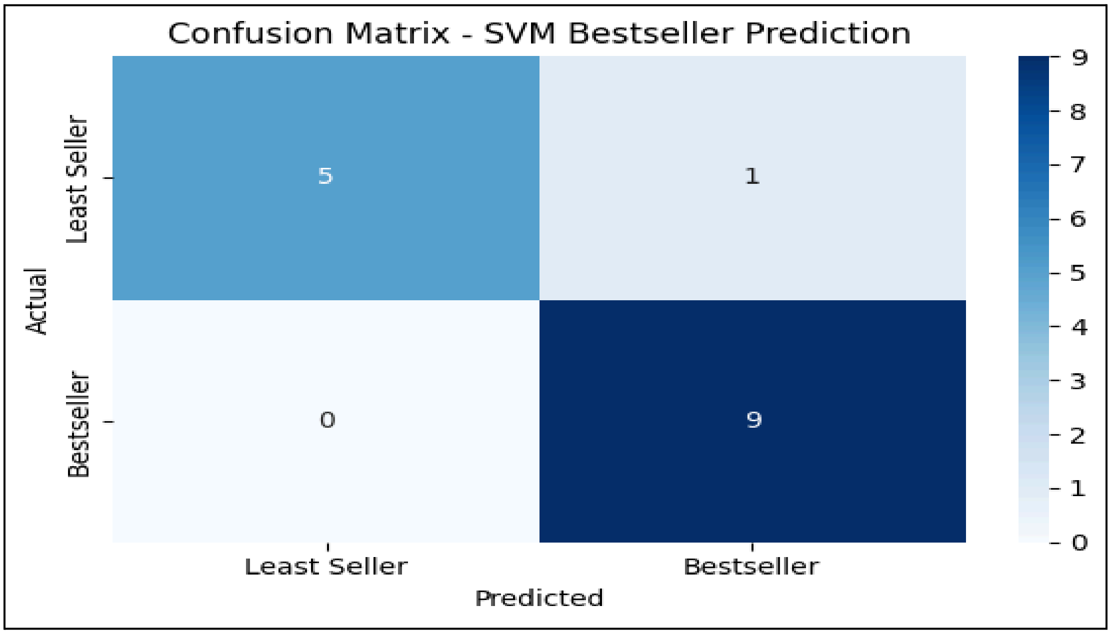

Figure 1.5.1 Support Vector Machine (SVM) Confusion Matrix 1

	Precision	**Recall**	**F1-Score**	**Support**
0	1.0	0.83	0.91	6
1	0.90	1.00	0.95	9
Accuracy			0.93	15
Macro Avg	0.95	0.92	0.93	15
Weighted avg	0.94	0.93	0.93	15

Table 1.5.1 Support Vector Machine (SVM) Classification Report 1

Fold 1:	0.93
Fold 2:	0.933
Fold 3:	1.0
Fold 4:	1.0
Fold 5:	1.0
Average Accuracy:	0.9733 or **97.33%**

Table 1.5.2 Support Vector Machine 5 Fold Cross-Validation Accuracy Scores 1

The KNN model achieved the highest accuracy (100%), making it exceptionally suited for accurately identifying best-selling items. However, due to scalability concerns, the SVM model, with an accuracy of 93%, might be more practical for larger datasets.

2. Customer Behavior

2.1 Decision Tree

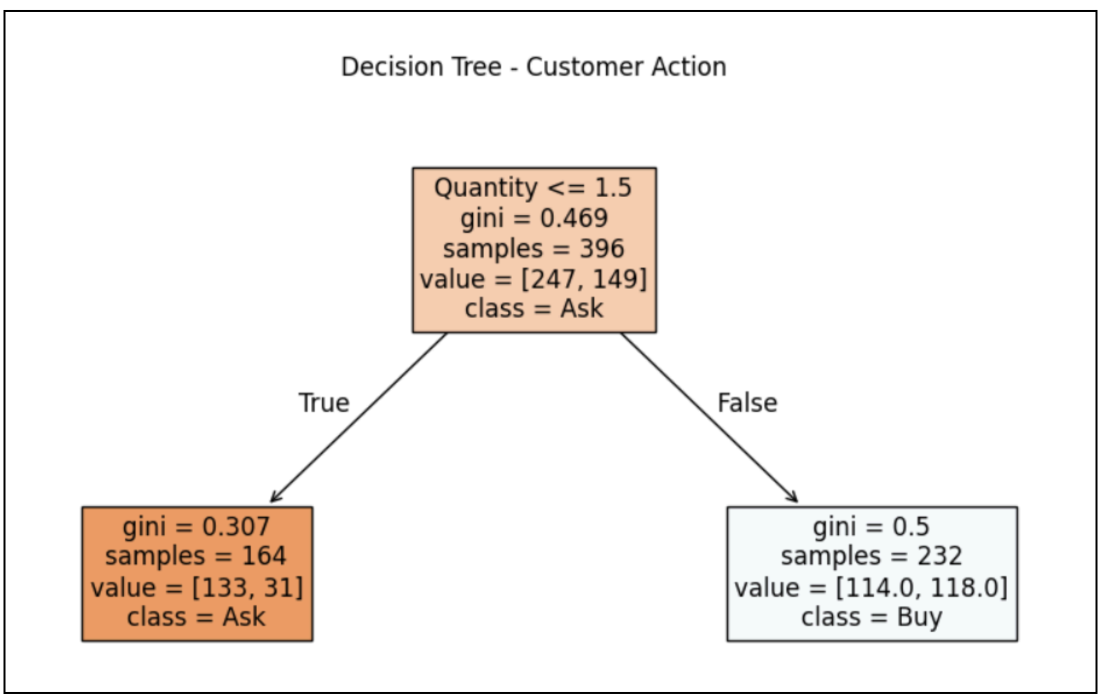

Figure 2.1.1 Decision Tree 2

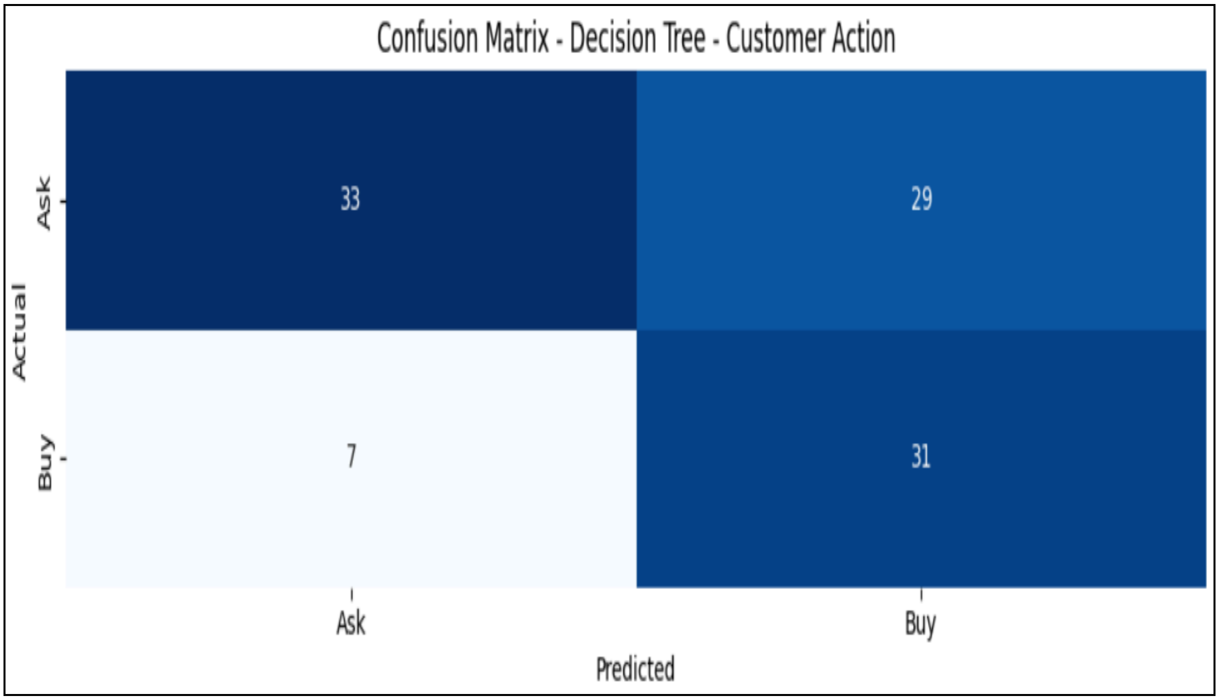

Figure 2.1.2 Decision Tree Confusion Matrix 1

	Precision	**Recall**	**F1- Score**	**Support**
0	0.82	0.53	0.65	62
1	0.52	0.82	0.63	38
Accuracy			0.67	70
Macro Avg	0.67	0.67	0.64	100
Weighted Avg	0.75	0.67	0.70	100

Table 2.1.1 Decision Tree Classification Report 2

Fold 1:	0.62
Fold 2:	0.61
Fold 3:	0.62
Fold 4:	0.36
Fold 5:	0.37

Average Accuracy:	0.64 or **64%**

Table 2.1.2 Decision Tree Fold Cross-Validation Accuracy Scores 2

2.2 Logistic Regression

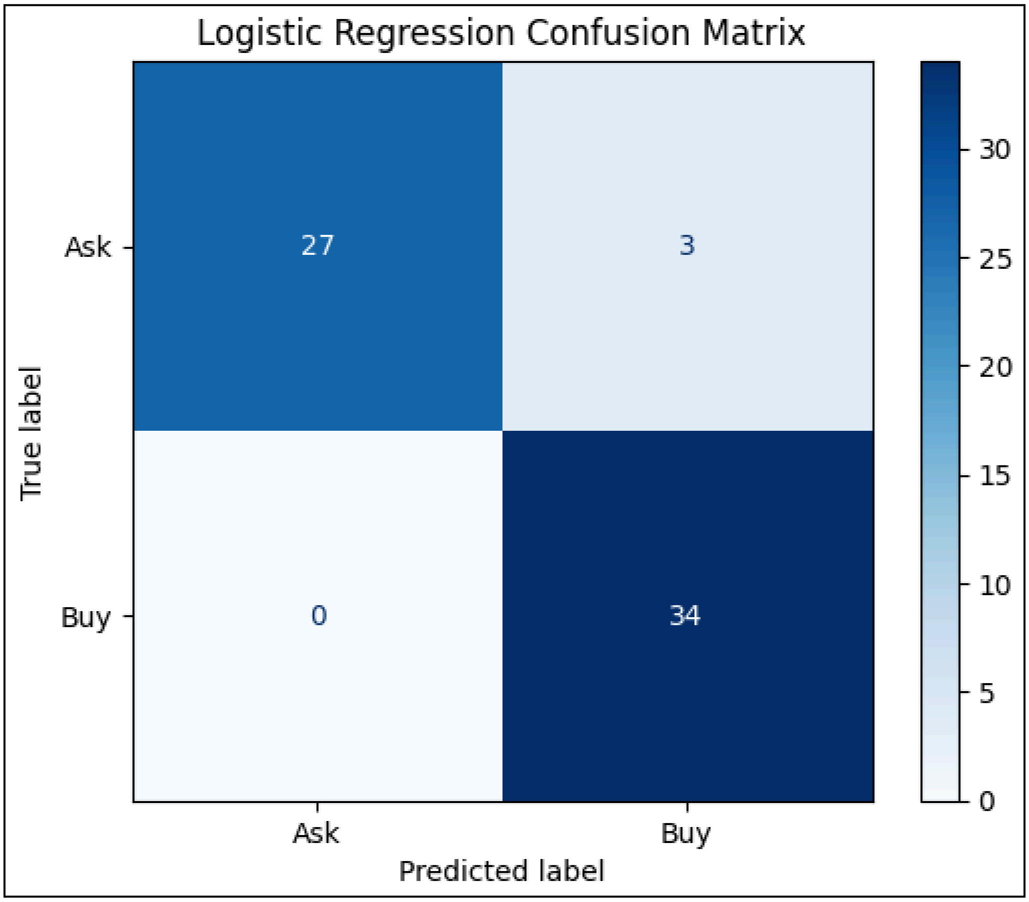

Figure 2.2.1 Logistic Regression Confusion Matrix 2

	Precision	**Recall**	**F1- Score**	**Support**
0	1.00	0.90	0.95	30
1	0.92	1.00	0.96	34
Accuracy			**0.95**	64

Macro Avg	0.96	0.95	0.95	64
Weighted Avg	0.96	0.95	0.95	64

Table 2.2.1 Logistic Regression Classification Report 2

Fold 1:	0.89
Fold 2:	0.92
Fold 3:	0.87
Fold 4:	0.90
Fold 5:	0.90
Average Accuracy:	0.90 or **90%**

Table 2.2.2 Logistic Regression 5 Fold Cross-Validation Accuracy Scores 2

2.3 KNN

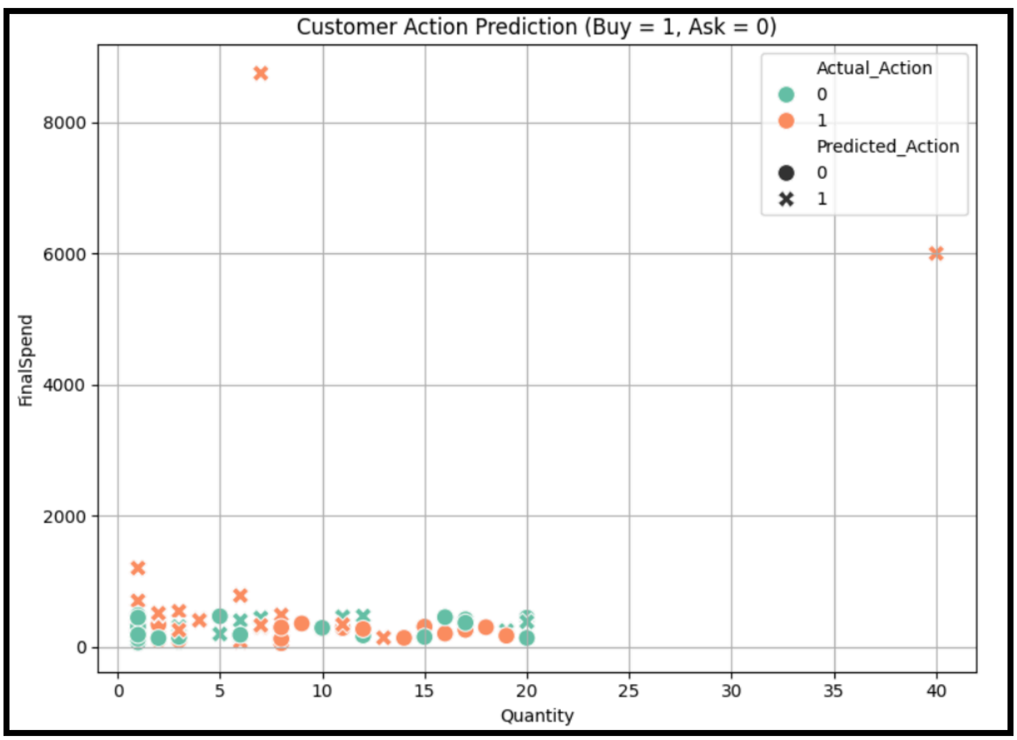

Figure 2.3.1 KNN 2

	Precision	Recall	F1- Score	Support
0	0.63	0.75	0.68	55
1	0.60	0.47	0.53	45
Accuracy			**0.62**	100
Macro Avg	0.62	0.61	0.60	100
Weighted Avg	0.62	0.62	0.61	100

Table 2.3.1 KNN Classification Report 2

Fold 1:	0.73
Fold 2:	0.78
Fold 3:	0.64
Fold 4:	0.57
Fold 5:	0.54
Average Accuracy:	0.65 or 65%

Table 2.3.2 KNN 5 Fold Cross-Validation Accuracy Scores 2

2.4 Naive Bayes

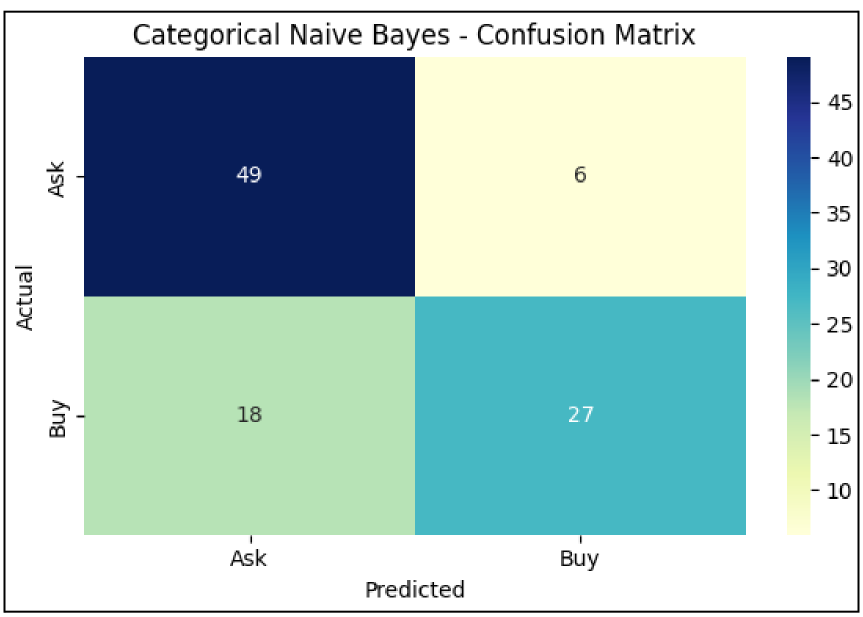

Figure 2.4.1 Naive Bayes Confusion Matrix 2

	Precision	**Recall**	**F1- Score**	**Support**
0	0.73	0.89	0.80	55
1	0.82	0.60	0.69	45
Accuracy			**0.76**	100
Macro Avg	0.77	0.75	0.75	100
Weighted Avg	0.77	0.76	0.75	100

Table 2.4.1 Naive Bayes Classification Report 2

Fold 1:	0.71
Fold 2:	0.78
Fold 3:	0.88
Fold 4:	0.72
Fold 5:	0.81
Average Accuracy:	0.78 or **78%**

Table 2.4.2 Naive Bayes 5 Fold Cross-Validation Accuracy Scores 2

2.5 Support Vector Machine (SVM)

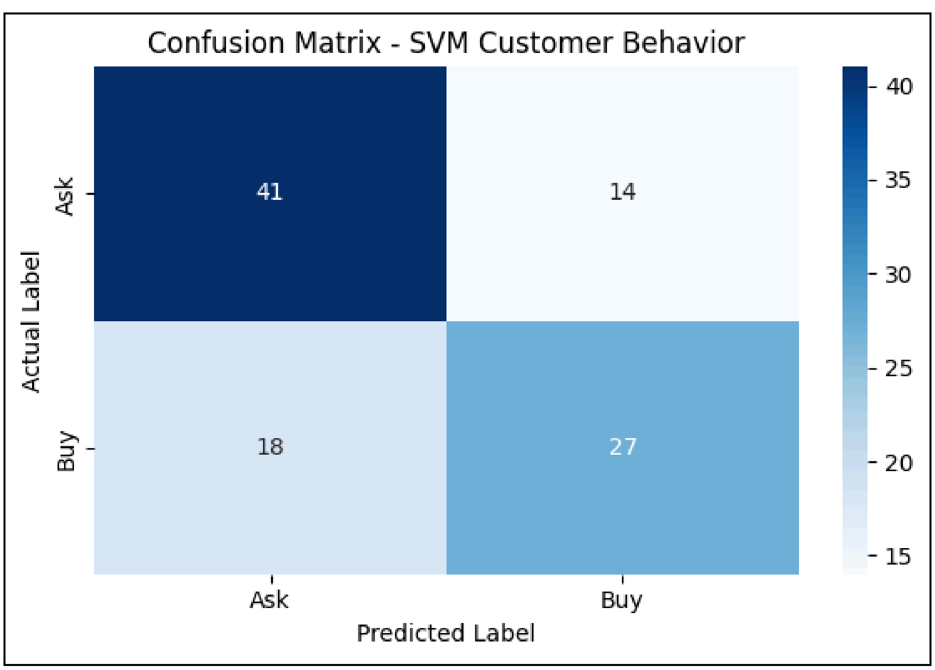

Figure 2.5.1 Support Vector Machine Confusion Matrix 2

	Precision	**Recall**	**F1-Score**	**Support**
0	0.69	0.75	0.72	55
1	0.66	1.60	0.63	45
Accuracy			0.68	100
Macro Avg	0.68	0.67	0.67	100
Weighted avg	0.68	0.68	0.68	100

Table 2.5.1 Support Vector Machine Classification Report 2

Fold 1:	0.72
Fold 2:	0.717
Fold 3:	0.626
Fold 4:	0.556
Fold 5:	0.596
Average Accuracy:	0.643 or **64.3%**

Table 2.5.2 Support Vector Machine 5 Fold Cross-Validation Accuracy Scores 2

Logistic Regression exhibited the best performance with a 95% accuracy rate, making it the most reliable model for predicting customer purchase intentions. The Naive Bayes model, though slightly less accurate (76%), is valuable for its simplicity and efficiency.

3. Predicting Monthly Restock Quantity

3.1 Linear Regression

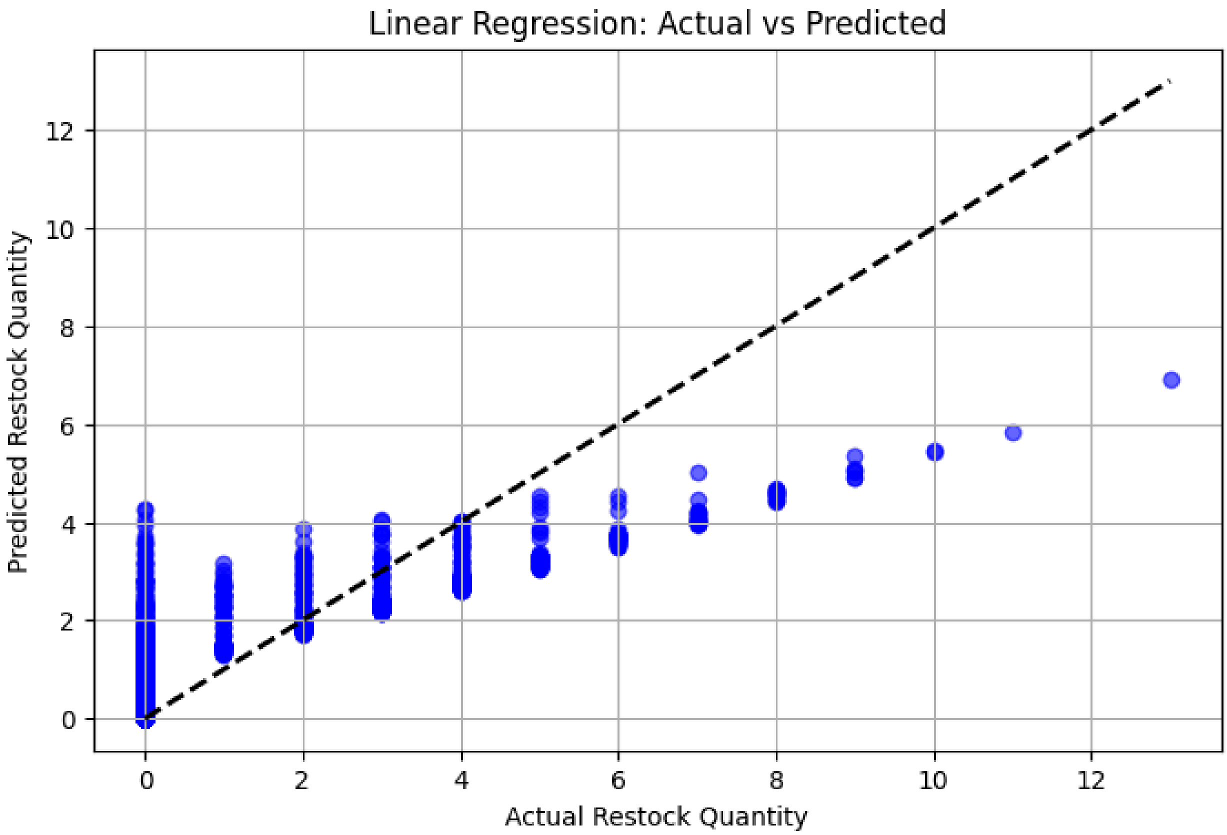

Figure 3.1 Linear Regression

Evaluation Metrics:

- **R^2 Score:** 0.6282902929960954
- **Root Mean Square Error (RMSE):** 1.2227018938147933
- **Mean Square Error (MSE):** 1.494999921138282
- **Mean Absolute Error (MAE):** 0.8129303990153558

3.2 Polynomial Regression

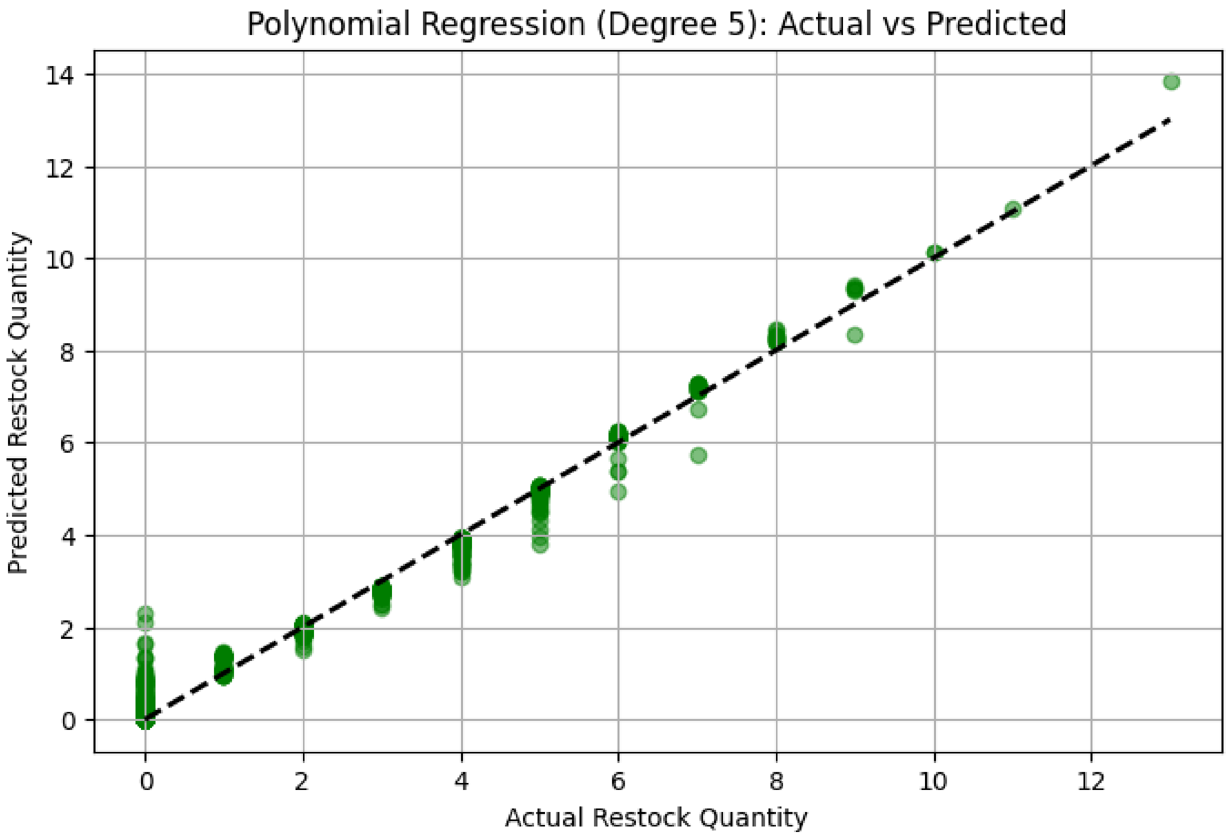

Figure 3.2 Polynomial Regression

Evaluation Metrics:

- **R² Score:** 0.9849042213130623
- **Root Mean Square Error (RMSE):** 0.24640322837853873
- **Mean Square Error (MSE):** 0.06071455095536632
- **Mean Absolute Error (MAE):** 0.1253405234025295

Polynomial Regression demonstrated outstanding predictive capability with an R^2 score of approximately 98.5%, significantly outperforming Linear Regression, which had an R^2 score of about 62.8%. Polynomial Regression is thus the preferred model for restocking predictions.

Classification Model Comparison: **Predicting Best and Least Selling Items**

Model	Task	Accuracy	Strength	Weakness
Logistic Regression	Classification	67%	The model performs well at identifying non-bestsellers, with 86% precision and 77% F1-score for class 0, showing it can effectively filter out items that are unlikely to become top sellers.	The model struggles to correctly identify actual bestsellers, achieving only 35% precision and 44% F1-score for class 1, meaning it frequently mislabels top-selling products
KNN	Classification	100%	Easy to implement, performs well on small, clean datasets; achieved 100% accuracy on the test.	Not scalable to large datasets, can overfit, sensitive to irrelevant or unscaled features.
Decision Tree	Classification	66.6%	Quick to analyze and gives clear insights on high-demand items.	Tends to overfit and may misjudge items close to the threshold, especially with limited or changing data.
SVM	Classification	93%	The 93% accuracy shows the model reliably identifies bestsellers, making it effective for inventory decisions.	It may still misclassify some low-selling items as bestsellers, which can lead to overstocking.
Naive Bayes	Classification	90%	The model delivers excellent overall performance with 90% accuracy, correctly identifying both best-selling and low-selling items with balanced precision and	Despite high accuracy, it still misclassified 10 items (6 low sellers as bestsellers and 4 vice versa), which might affect inventory decisions

			recall.	if left unchecked.

Classification Model Comparison: ***Predicting Customer Behavior***

Model	Task	Accuracy	Strength	Weakness
Logistic Regression	Classification	95%	The logistic regression model demonstrates strong predictive performance, achieving 95% overall accuracy with perfect precision for class "Ask" and perfect recall for class "Buy", indicating it can confidently identify buyers without falsely labeling non-buyers.	The model misclassified 3 out of 30 "Ask" cases as "Buy", suggesting it may sometimes overestimate buyer intent, which could lead to wasted marketing efforts in customer engagement.
KNN	Classification	62%	Easy to use and interpretable; performs moderately well at identifying customer intent (especially those who only ask).	Struggles to correctly predict buyers; lower recall for class 1; may require more features for better prediction.
Decision Tree	Classification	64%	Simple and understandable, reasonably good at spotting customers who only inquire.	Has difficulty identifying actual buyers, needs more data features to improve accuracy.
SVM	Classification	68%	The model captures general patterns in customer behavior, correctly predicting many "Buy" actions even with limited features.	The 68% accuracy suggests it struggles to distinguish "Ask" vs "Buy" in borderline cases, possibly due to overlapping spending and quantity patterns.

Naive Bayes	Classification	76%	The model shows strong performance in identifying "Ask" actions with a high recall of 0.89, meaning it correctly detects most non-buying customers.	It struggles with predicting actual "Buy" actions, misclassifying 18 out of 45 buyers, which lowers its reliability for purchase prediction.

Regression Model Comparison: ***Predicting Stock Quantity Needed Monthly***

Model	Task	R^2 Score	RMSE	Strength	Weakness
Linear Regression	Regression	62.8%	1.22	Linear regression is simple and fast to train. It's good for baseline models or data with linear trends.	However, it cannot capture curves or complex patterns. It's also not flexible for seasonal or non-linear behavior.
Polynomial Regression	Regression	98.5%	0.25	Polynomial regression can model curved relationships, capture non-linear trends, and perform much better on complex datasets.	However, it can overfit if the degree is too high, it's harder to interpret than linear, and it's sensitive to outliers and unscaled data.

Superiority:

Logistic Regression and SVM generally yield balanced accuracy, interpretability, and scalability. Polynomial Regression consistently outperforms Linear Regression in complex forecasting tasks.

Scale of Errors:

Regression models were evaluated using R^2 Score and Root Mean Square Error (RMSE). Classification models used accuracy, precision, recall, and F1-score. Polynomial Regression had the lowest error margin (RMSE ≈ 0.25), indicating strong predictive reliability for inventory estimation.

Model Complexity vs Performance:

- **Low complexity, moderate accuracy**: Linear Regression, Logistic Regression, Naive Bayes
- **Moderate complexity, high accuracy**: SVM, Decision Tree
- **High complexity, high accuracy**: Polynomial Regression

A trade-off exists between interpretability and precision, particularly with non-linear models.

Implications for Model Selection:

- **Inventory Forecasting**: Polynomial Regression is ideal due to high predictive accuracy for restocking quantities.
- **Customer Action Prediction**: SVM or Decision Tree models are recommended for real-time decision-making.
- **Operational Simplicity**: Naive Bayes or Logistic Regression can be deployed with minimal computing overhead.

Summary of All Models Forecasted Values:

Regression models provided numerical estimations for stock quantity needs according to the item characteristics, such as price, current stock levels, and units sold per month. Classification models identified whether customers were likely to purchase, and which products would trend as bestsellers.

Key Insights:

- High-spending and large-quantity transactions are strong predictors of bestsellers.

- Logistic Regression is best for predicting customer intent, like buy vs ask.
- SVM and Decision Tree models provided a high classification accuracy for best sellers and customer behavior analysis.
- Manual estimates led to mismatches in inventory that are now corrected with predictive modeling.
- Polynomial Regression excels in complex patterns, ideal for monthly inventory forecasting.

Conclusion:

By integrating machine learning into its operations, Fukushimart has transitioned from reactive decision-making to a predictive, data-driven approach. In classifying best selling items, the K-Nearest Neighbors (KNN) model achieved the highest performance with a perfect accuracy of 100%, though it may not scale well with larger datasets. Support Vector Machine (SVM) followed with a strong accuracy of 97.33%, making it highly effective and scalable for real-time inventory classification. For predicting customer purchase behavior, Logistic Regression stood out with an impressive accuracy of 95%, making it the most reliable model for identifying likely buyers. Naive Bayes also performed well with an accuracy of 76%, offering simplicity and efficiency. In the regression task of forecasting monthly stock quantity, Polynomial Regression achieved an outstanding R^2 score of 98.5% with a low RMSE of 0.25, outperforming Linear Regression which had a 62.8% R^2 score and a higher RMSE of 1.22. These results clearly demonstrate that machine learning models not only improve inventory planning and customer targeting but also enhance overall operational efficiency. For practical deployment, it is recommended to use SVM for customer behavior classification, Polynomial Regression for stock forecasting, and Logistic Regression for quick, interpretable predictions.

Recommendation:

- Deploy SVM for real-time purchase classification to improve customer targeting.
- Use Polynomial Regression for monthly inventory planning and restock prediction.
- Regularly retrain models with new transaction data to maintain accuracy.
- Integrate these models into the existing virtual store dashboard for streamlined operations.